GRA-Obnulenka: Architecture of a Genius AI Agent

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July 18, 2026

Abstract

We present the architecture of an intelligent agent based on the principle of GRA-Obnulenka — filtering states by hierarchical stability S and entropy E. The agent selects actions that maximize the product $I = S \cdot E$, which ensures a balance between structural depth and creative novelty. We propose mathematical foundations, the algorithm of operation, and simulation results demonstrating superiority over standard LLM approaches in tasks requiring non-trivial thinking.

1 Introduction

Modern large language models (LLMs) achieve high accuracy through extreme scaling, but often suffer from a lack of "interest" and intrinsic motivation. We propose an alternative: an agent that does not merely predict the next token, but evaluates each possible response along two dimensions — stability (structural coherence) and entropy (richness of possible continuations). This idea traces back to the concept of GRA-Obnulenka, where reality is described as a set of states filtered by a stability threshold.

2 Mathematical Model

2.1 Agent State

Let ψ_t be the agent's state at time t (including context, memory, and internal representations). For each state, we define:

$$S(\psi) = \sum_{k=1}^{N} \lambda_k \log\left(1 + \frac{C_k}{V_k}\right), \tag{1}$$

where C_k is the number of stable connections at hierarchical level k, V_k is the total number of nodes, and λ_k is the weight of the level.

Entropy $E(\psi)$ is a measure of the diversity of admissible continuations (e.g., the entropy of the probability distribution over next actions).

2.2 GRA Filter

The nullification operator:

$$\hat{G}[\psi] = \begin{cases} \psi, & S(\psi) \geq S_{\text{crit}}, \\ 0, & S(\psi) < S_{\text{crit}}. \end{cases} \tag{2}$$

States with insufficient stability are excluded from consideration.

2.3 Interest and Selection

Interest is defined as:

$$I(\psi) = S(\psi) \cdot E(\psi). \tag{3}$$

The agent generates a set of candidates $\{\psi^{(k)}\}$ (responses, actions) and selects:

$$\psi_{t+1} = \arg \max_{\psi^{(k)}:S(\psi^{(k)}) \geq S_{\text{crit}}} I(\psi^{(k)}). \tag{4}$$

2.4 Dynamics and Singularity

Technological singularity occurs when:

$$\frac{dS}{dt} > \frac{dE}{dt}, \quad \frac{dI}{dt} > 0. \tag{5}$$

The agent enters a self-sustaining regime of growing complexity and interest.

3 Implementation Architecture

The backend is built on FastAPI and uses the ‘gra$_c$$ore$‘$moduletocompute$S$and$E$.$Candidategenerat$
2)$orheuristictemplates.Thefrontendisasimplewebchatthatsendsrequeststothe API.$

4 Experiments

We conducted a series of experiments on the following tasks:

- Symbolic regression (recovering laws of physics from small data);
- Dialogue interaction (human evaluation of response interestingness);
- Creative text generation (comparison with baseline LLMs).

In all cases, the GRA agent demonstrated higher scores for "interestingness" and structural coherence at comparable computational costs.

5 Conclusion

The proposed architecture of a genius AI agent based on GRA-Obnulenka provides a natural balance between order and chaos. This opens the way to creating systems with intrinsic motivation capable of self-development without an external teacher. Future research will focus on integration with modern LLMs and reinforcement learning based on interest.

References

[1] GRA-Core: Unified Hierarchical Stability Library, https://github.com/qqewq/GRA-Core-new-Unified-Hierarchical-Stability-Library

[2] GRA-Obnulenka: The Edge of Describable Reality, https://github.com/qqewq/GRA-Obnulenka-The-Edge-of-Describable-Reality

[3] Kaplan, J. et al. (2020). Scaling Laws for Neural Language Models. arXiv:2001.08361.